Provenance-Preserving Structure-Aware Chunking for Auditable Enterprise Retrieval-Augmented Generation

***Abstract*-Enterprise retrieval-augmented generation requires evidence that is both relevant and attributable to its source, yet conventional chunkers can separate a requirement from its constraints, a table row from its header, or an interface field from its section context. We present SEAC, a deterministic structure-aware chunker that constructs provenance-carrying document units, protects structural groups, combines semantic, structural, and key-information-density signals, and indexes table rows as precise children of coherent parent chunks. A frozen study covers 66,981 units from ten enterprise software projects and compares 15 methods using BGE-M3 embeddings. On six retrieval-eligible locked projects and 125 human-approved questions, SEAC obtains 0.7937 unit MRR and 0.9520 Hit@5; the strongest non-SEAC baseline obtains 0.7290 MRR. The prespecified comparison with LlamaIndex yields a project-paired difference of 0.1003 with exact p = 0.0625, so the evidence does not establish a statistically significant confirmatory advantage. Boundary F1 is 0.5673 on six independently eligible projects. Removing table-row children reduces MRR to 0.6853, whereas removing parent fusion increases it to 0.8289. SEAC therefore improves observed evidence retrieval and auditability at higher indexing cost, but does not by itself establish safer generated answers.**

Keywords-retrieval-augmented generation, document chunking, provenance, auditability, software engineering, trustworthy artificial intelligence

I. INTRODUCTION

Retrieval-augmented generation (RAG) grounds a language model in external evidence [1]. In enterprise software engineering, that evidence is distributed across requirements, interface specifications, database dictionaries, detailed designs, technical agreements, issue records, and meeting minutes. The relevant fact is often not an isolated sentence. A requirement carries force through an obligation term and its section; an interface parameter depends on its endpoint; and a database value is meaningful under a column header and constraint. A flat character window can retrieve plausible words while losing exactly the context needed to audit an answer.

This creates a transparency problem before generation begins. If retrieval returns a passage whose source units are unknown, or a table whose correct row cannot be distinguished, a downstream system cannot provide a defensible citation. Conversely, returning an entire document increases irrelevant context and can hide the supporting span. Chunking should therefore balance coherent returned context, fine-grained matching, and stable provenance.

We propose SEAC, a deterministic structure- and key-information-aware adaptive chunker for auditable enterprise RAG. SEAC parses ordered units with stable source identifiers, prevents cuts through protected groups, uses adjacent semantic distance together with explicit document signals, and creates retrieval children whose matches resolve back to a coherent parent. Table rows retain their headers and original row identity. Every returned parent carries its ordered source-unit lineage.

This paper makes three contributions. First, it defines a provenance-preserving chunk representation and a transparent boundary rule for heterogeneous engineering documents. Second, it provides a frozen industrial evaluation with human approval and privacy-release gates, axis-specific withholding rules, and project-clustered inference. Third, it reports both quality and cost, including negative ablation findings. The evaluation measures evidence retrieval and boundary agreement; it does not claim that chunking alone prevents hallucination, prompt injection, or privacy leakage.

II. RELATED WORK

RAG and dense passage retrieval connect a query to external text before generation [1], [2]. Retrieval granularity is consequential: smaller units improve localization but can lose context, while larger units retain context but dilute the matching signal [4]. Recent work explores long-context embeddings, recursive hierarchies, learned chunk mixtures, and systematic RAG design [5]-[8]. Document segmentation itself materially affects retrieval quality [6], yet most evaluations use public prose or synthetic transformations rather than confidential software artifacts with tables and repeated specifications.

Document conversion systems can recover headings, reading order, and tables [9], but extraction does not determine which elements should share a retrieval unit. Software traceability research links requirements to downstream artifacts [10], [11]; it motivates stable source identity and evidence equivalence when the same approved passage occurs in several documents. Long-context studies further show that merely placing evidence in a large context does not guarantee effective use [12]. SEAC focuses on the missing interface between structured document conversion and provenance-aware enterprise retrieval.

III. SEAC

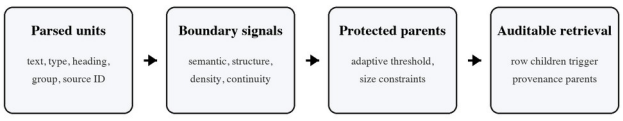


Fig. 1. SEAC preserves stable source identifiers through boundary scoring, parent construction, and child-triggered retrieval.

A. Provenance-Carrying Units

Let a parsed document be an ordered sequence of units (D = (u_1, ..., u_n)). Each unit stores text, a stable unit identifier, its source block, document type, block type, heading path, and an optional protected-group identifier. Paragraphs and headings are units. For a table with header (h), each row (r_j) is rendered as

$$t(r_j) = header(h) \,||\, row(r_j)$$

so a row remains interpretable outside the table. A parent chunk stores the ordered unit and block identifiers from which it was assembled. Retrieval correctness can then be resolved by identifier rather than by approximate string matching.

B. Transparent Boundary Scoring

For adjacent units, semantic distance is (d_i = 1 - cos(e_i,e_i+1)), where (e_i) is a normalized dense embedding. A structural score (s_i) combines a new-heading indicator, heading-path change, and protected-group transition. A frozen engineering lexicon counts obligation, constraint, interface, field, table, process, and limit terms. Its normalized density is

$$rho(u) = sum(v \text{ in } V)\ count(v,u) / max(1, tokens(u))$$

and the adjacent density-change score is (k_i = min(1,10| rho(u_i)-rho(u_i+1)|)). Continuity penalty (p_i) discourages cuts within a heading path and forbids practical separation inside a protected group. The frozen score is

$$b_i = 0.45\ d_i + 0.30\ s_i + 0.25\ k_i - p_i$$

For document (D), the threshold is (theta_D = Q_0.8(b) + 0.08r_t - 0.05r_h), where (r_t) and (r_h) are the fractions of table rows and headings. Parent construction is a deterministic left-to-right pass with a 512-token hard maximum, a 320-token ordinary target, a 224-token target in dense regions, and a 48-token minimum. Candidate cuts are suppressed when they split a protected group or detach a heading from its first content unit.

C. Hierarchical Retrieval

Each parent is the context returned downstream. Non-heading, non-noise units also produce retrieval children with a 112-token target and 176-token maximum. Table rows normally become individual children and retain row provenance. Retrieval text is prefixed with document type and heading path. For parent score (S_p) and child scores (S_c), full SEAC ranks

$$S(p,q) = 0.35\ S_p(q) + 0.65 \max S_c(q)$$

A precise row can therefore activate its parent while the returned text preserves nearby constraints. Excluding encoder cost, construction is linear in the number of units; index size grows with parents plus children.

IV. INDUSTRIAL EVALUATION

A. Corpus and Frozen Protocol

The corpus contains Chinese-language DOCX and XLSX artifacts from ten anonymized enterprise projects. It includes requirements, API specifications, database designs, detailed designs, technical agreements, meeting minutes, proposals, and research reports. Six unsupported PDF or PPTX files are disclosed as substantive skips. Table I summarizes the fixed split. P01-P02 are development projects, P03 is validation, and P04-P10 are collectively locked. No locked result may change parameters, question wording, baseline selection, or exclusion rules.

TABLE I. AGGREGATED INDUSTRIAL CORPUS

Projects	Documents	Units	Regex tokens
2	15	21,371	418,541
1	7	5,193	103,562
7	67	40,417	1,155,797
10	89	66,981	1,677,900

All methods use BGE-M3 [3], a 512-token maximum, and unit-level evidence. We compare fixed windows, a local recursive splitter, structure-only and semantic splitters, official LangChain and LlamaIndex splitters, non-hierarchical SEAC, full SEAC, and seven single-component ablations. The formal matrix contains 15 methods.

B. Human and Privacy Gates

Boundary references are labelled BREAK, NO_BREAK, or NOISE. P01 has 4,783 independently double-labelled and adjudicated boundaries. For P02-P10, 300 stratified candidates per project were pre-labelled by Claude Opus 5 Thinking through Cursor under a frozen rule set and without method outputs. A human then labelled a blinded random sample of 100 per project plus every case the model flagged. Human labels override pre-labels. A project requires Cohen's kappa of at least 0.70 [13]. P08 reached 0.650 and was withheld only from boundary evaluation; retrieval remains eligible. Across promoted references, 77.5% of 7,183 candidates are human verified, 5.4% rule determined, and 17.1% retain the model pre-label.

Question drafts undergo mechanical evidence checks, advisory model pre-review, and human approval. Evidence is a set of stable unit identifiers because approved spans can legitimately repeat across documents. Questions with ambiguous rows or non-distinctive spans are rejected. Privacy status is assigned only by an authorized human; records are admitted only when marked no-risk, anonymized, or explicitly cleared. P10 retained 11 approved questions, below the frozen minimum of 15, and is withheld only from retrieval evaluation. Consequently, the locked analyses contain 125 questions from six retrieval-eligible projects and 1,368 candidates from six boundary-eligible projects.

C. Metrics and Inference

Unit MRR and Hit at (k) require a returned parent's provenance to contain an approved evidence unit. Span metrics additionally check the minimal supporting text. Boundary metrics compare adjacent parent decisions after excluding NOISE. We form 95% intervals with 10,000 hierarchical bootstrap samples, resampling projects and then items [14]. The single confirmatory test compares SEAC with LlamaIndex SentenceSplitter on unit MRR. It exchanges all matched observations within a project together and enumerates all 64 assignments across six projects. Secondary retrieval and boundary families use Holm correction [15].

V. RESULTS

A. Retrieval and Boundary Quality

Table II reports locked-test results. SEAC reaches 0.7937 unit MRR and 0.9520 Hit@5. The strongest non-SEAC retrieval baseline is structure-only at 0.7290 MRR, an observed absolute difference of 0.0647. SEAC also yields 0.8561 span MRR. Against the prespecified LlamaIndex comparator, the project-paired MRR difference is 0.1003 with bootstrap interval [0.0042, 0.2008] and exact two-sided p = 0.0625. SEAC is higher on five of six retrieval projects, but the confirmatory null is not rejected at 0.05.

TABLE II. LOCKED-TEST RESULTS

	Unit MRR	Hit@5	Span MRR
	0.5158	0.7200	0.1350
	0.6086	0.8080	0.7072
	0.7290	0.8880	0.8211
	0.6999	0.8640	0.8121
	0.6460	0.8480	0.7329
	0.6934	0.8880	0.7586
	0.6796	0.8800	0.7770
	0.7937	0.9520	0.8561

SEAC boundary precision is 0.4321, recall is 0.8255, and F1 is 0.5673 with interval [0.4708, 0.6488]. Structure-only is the strongest non-SEAC boundary baseline at 0.5032. The paired F1 difference is 0.0640 with interval [0.0313, 0.0956], but the Holm-adjusted p value is 0.4375. The high recall and moderate precision indicate that SEAC favors coherent separation at the cost of additional boundaries.

B. Ablations and Cost

Table III prevents a one-sided component story. Removing table-row children produces the largest observed degradation, reducing MRR by 0.1084 and Hit@5 from 0.9520 to 0.8640. However, the Holm-adjusted secondary p value for this comparison is 1.0000. Removing semantic distance changes MRR by less than four decimal places, and removing parent fusion increases MRR to 0.8289. Heading context also increases MRR when removed, although Hit@5 falls to 0.9280. Thus, row-level evidence is useful in this benchmark, while the fixed parent-fusion and heading-prefix choices are not supported as universally beneficial.

TABLE III. SINGLE-COMPONENT ABLATIONS

	Unit MRR	Hit@5	
	0.7937	0.9520	
	0.7937	0.9520	
	0.7959	0.9520	
	0.7979	0.9520	
	0.7959	0.9520	
	0.8129	0.9280	
	0.6853	0.8640	
	0.8289	0.9520	

SEAC creates 8,746 parents and 48,912 retrieval vectors, 5.50 times the vector count of structure-only. Aggregate locked-set chunking and indexing take 112.11 s and 351.12 s. Mean query latency is 20.93 ms versus 15.18 ms for structure-only, a 5.74 ms increase. This is an explicit auditability-quality-cost tradeoff rather than a free improvement.

VI. DISCUSSION AND LIMITATIONS

The results support a bounded claim: preserving tables and explicit source lineage can improve observed evidence retrieval in heterogeneous enterprise documents. Unit-level and span-level metrics answer different audit questions. A span match shows that relevant text appeared; a unit match shows that it came from an approved source location. This distinction matters when requirements are copied across proposals and specifications or when several table rows share the same vocabulary.

SEAC is also a transparent component, not an end-to-end safety guarantee. The study does not measure generated-answer faithfulness, adversarial prompts, access-control enforcement, or privacy leakage. Confidential inputs remain protected by external authorization and release gates. The raw corpus cannot be public; an anonymous artifact can include code, frozen configurations, schemas, synthetic examples, hashes, and aggregate analysis. Controlled access to additional artifacts depends on data-owner approval.

Validity is limited by ten projects from one industrial setting, six eligible project clusters per evaluation axis, Chinese documents, one embedding model, and unsupported image-heavy files. The mixed boundary reference includes model pre-labels, although its source is recorded per row and low-agreement P08 is excluded. Post-hoc clarification of three noise categories is disclosed, and agreement is retained before and after the clarification. Finally, higher MRR without confirmatory significance and the negative fusion ablation both argue for evaluation on additional independent projects rather than a general superiority claim.

VII. CONCLUSION

SEAC combines provenance-carrying units, protected structure, transparent boundary signals, and row-level children for auditable enterprise RAG. In the locked study it obtains 0.7937 unit MRR and 0.9520 Hit@5, with an observed but non-significant confirmatory advantage over LlamaIndex. Table-row children are the clearest useful component, while parent fusion and heading context require redesign. Future work should evaluate end-to-end answer faithfulness, adversarial robustness, multilingual public corpora, and learned cost-aware fusion under independently locked projects.

ACKNOWLEDGMENT

Funding and identity-bearing acknowledgments are withheld for double-blind review. Anthropic Claude Opus 5 Thinking was used as a recorded boundary pre-labeling instrument as described in Section IV-B. OpenAI Codex assisted code drafting, test generation, manuscript structuring, and language editing. Human reviewers overrode experimental pre-labels, verified the analyses and text, and retain responsibility for all research and publication decisions.

REFERENCES

[1] [1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Kuttler, M. Lewis, W. Yih, T. Rocktaschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Advances in Neural Information Processing Systems, vol. 33, 2020.

[2] [2] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. Yih, "Dense passage retrieval for open-domain question answering," in Proc. EMNLP, 2020, pp. 6769-6781.

[3] [3] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, "M3-Embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation," in Findings of ACL, 2024, pp. 2318-2335.

[4] [4] T. Chen, H. Wang, S. Chen, W. Yu, K. Ma, X. Zhao, H. Zhang, and D. Yu, "Dense X retrieval: What retrieval granularity should we use?" in Proc. EMNLP, 2024, pp. 15159-15177.

[5] [5] X. Wang et al., "Searching for best practices in retrieval-augmented generation," in Proc. EMNLP, 2024, pp. 17716-17736.

[6] [6] Z. Wang et al., "Document segmentation matters for retrieval-augmented generation," in Findings of ACL, 2025, pp. 8063-8075.

[7] [7] J. Zhao et al., "MoC: Mixtures of text chunking learners for retrieval-augmented generation system," in Proc. ACL, 2025, pp. 5172-5189.

[8] [8] P. Sarthi, S. Abdullah, A. Tuli, S. Khanna, A. Goldie, and C. D. Manning, "RAPTOR: Recursive abstractive processing for tree-organized retrieval," in Proc. ICLR, 2024.

[9] [9] C. Auer et al., "Docling technical report," arXiv:2408.09869, 2024.

[10] [10] J. L. C. Guo, J. Steghofer, A. Vogelsang, and J. Cleland-Huang, "Natural language processing for requirements traceability," arXiv:2405.10845, 2024.

[11] [11] S. J. Ali, V. Naganathan, and D. Bork, "Establishing traceability between natural language requirements and software artifacts by combining RAG and LLMs," in Conceptual Modeling, 2024, pp. 295-314.

[12] [12] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the middle: How language models use long contexts," Trans. Assoc. Comput. Linguistics, vol. 12, pp. 157-173, 2024.

[13] [13] J. Cohen, "A coefficient of agreement for nominal scales," Educ. Psychol. Meas., vol. 20, no. 1, pp. 37-46, 1960.

[14] [14] B. Efron, "Bootstrap methods: Another look at the jackknife," Ann. Stat., vol. 7, no. 1, pp. 1-26, 1979.

[15] [15] S. Holm, "A simple sequentially rejective multiple test procedure," Scand. J. Stat., vol. 6, no. 2, pp. 65-70, 1979.

[16] [16] LangChain, "RecursiveCharacterTextSplitter, langchain-text-splitters 1.1.2," 2026. [Online]. Available: https://pypi.org/project/langchain-text-splitters/1.1.2/

[17] [17] LlamaIndex, "SentenceSplitter, llama-index-core 0.14.23," 2026. [Online]. Available: https://pypi.org/project/llama-index-core/0.14.23/